

Multilevel models (W02)

Example question

Q: Which subject-level estimates get shrunk the most?

A: <add your answer>

Q: how is multiple regression with two predictors different from multilevel modelling with two predictors?

A: Multilevel models need random effects on top of the fixed level effects. The difference between using the `lm` and `glmer` function in R.

Q: Explain `Corr` and what it means / outcome?

A: function for correlation between the arguments you put in

Q: How is the variance-covariance connected to `G` and what does it actually do?

***A:** `G` is an estimated ratio describing the variance-covariance matrix of the second level / random effects. It expresses the random-effect variances relative to the residual noise. `G` transforms the standard deviations of residuals to the covariance matrix for the intercept and slope coefficients. For example, 0.93 means the between-subject intercept variance is about 93% as large as the within-subject residual variance.

- 0.93 (intercepts): People differ in their baseline reaction time by almost as much as the random noise. Who you are matters about as much as random variation.
- 0.054 (Days slopes): People also differ in how fast they get worse, but that difference is small compared to the noise.
- 0.015 (covariance): Being slow at the start barely tells you anything about how quickly someone gets worse.

***We looked at all of the lecture** notes and tried to understand where `G` comes from but still don't quite understand. It would be helpful to know how you get `G` and how to interpret it :)

Q: What does the different pooling mean?

A: 3 kinds of pooling, no pooling, partial pooling and complete pooling. No pooling: Treats subjects like they do not share ... and overstates variation among them. Complete pooling: Ignores random effects (subject specific differences), the same as using the `lm` function in R. Partial pooling: Includes random effects (subject level differences) and balances the dataset so extreme cases (subjects with fewer data points) gets pulled towards the mean.

Q: Why is there little shrinking in partial pooling?

A: we take into account how many subjects were included in the group, so a group with many subjects would explain way more than a group with almost no subjects, so the group with few subjects gets penalized and shrunk

Q: What does it mean for the model that each subject get an individual intercept/slope? Does it mean that we get a lot of different lines if we were to visualize it?

A: Yes, you'd get a lot of different lines. If you only add separate intercepts, then each line starts at different heights on the y-axis, but the angle of the lines are the same. If you also add the slopes, then the angle also is different for each line.

Q: What does it mean that G can't be inverted if there is too much covariance?

A: MATH, Because the correlation between the two is big, they can't actually be separated, therefore it causes them to be on the same regression line.

Q: How will this look visually? -One intercept per subject with four slopes from there (one per PAS)
`slope_pas_sub <- glmer(count ~ pas * task + (pas - 1 | subject), data=data.count, family = "poisson")`

A: we have count on the y-axis, and we can have pas the x-axis, and then we can group by tasks, so there would be four slopes for each task, one slope for the difference in estimated count between PAS1 and PAS2, another slope between PAS2 and PAS3, etc.

Q: What does it mean that the random variable Y in eq. 2, is conditional on the instantiation, b, of the random variable B?

A:

Generalised linear modelling (W03)

Example question

Q: What is a link function?

A: A link function is a function that serves to "translate" nominal, ordinal or interval data into ratio data, which can then be subjected to further data analysis.

Please add more questions and answers:

Q: When you use a link function to make your model, how do you undo the transformation when you want to get your predicted values/ $\hat{y}$'s?

A: take the inverse of the inverse to get it back to "normal"

Q: Why do we have to transform binomial data with the logit function?

A: Very basically, because binomial data on the original scale isn't linear, and we therefore can't fit a linear model (even if multi-model) over it. So, you use a link function to transform your data into a scale where you can fit a linear line over it. For binomial data, you'd logit transform it.

Q: when and why do we need a link function g as an "ingredient" for Poisson and logistic regression?

A: In order to fit the data of a non-linear model you need the link function to conform it into a generalized linear model. Log transform data for Poisson and log transform for logistic(binomial).

Q: if you plot design matrix X and Z in R with `image.plot` what do you actually see and why?

A: You see the values of the betas when plotting an X matrix. And the colours correspond to the values (you have the scale on the right of the plot). When modelling the Z matrix, the colours of the plot correspond to the values of β s (so kinda betas of the Z matrix)

Q: What is a link function and why do we do it?

A: we take the inverse of the function to make it linear. So now we get probability between 0 and 1

Q: When you model a model with `glmer(formula = y ~ x1 * x2 + (0 | x1 + x2))`, with x1 and x2 being ategorical predictors, you are not modeling a unique slope for x1, but what are you modeling instead?

A: you are modeling the slope between the different categorical predictors in x1 for each of the different x2

Evaluating generalised multilevel models (W04)

Example question

Q: What is being compared in a likelihood-ratio test? Which values do you compare/look at in the output?

A: <add your answer>

Please add more questions and answers:

Q: What is the AIC?

A: Akaike Information Criteria. A way to evaluate the goodness of a model. Calculated like $AIC = deviance + 2k$, where deviance is $-2(\log_likelihood)$ and k is number of parameters. The lower the AIC, the better the model (typically).

Q: What does it mean when a model has a converging fit? What does it mean when a model has a singular fit? Is that the same as when a model 'failed to converge'?

A:

Q: how do we estimate sigma and alpha with a graph of a staircase formation? [slides p.21]

A:

Q: How do you interpret the Chi squared value in the ANOVA, and what makes a significant log likelihood according to it?

A:

Q: What is the purpose and difference between $R^2(m)$ and $R^2(c)$?

A: $R^2(m)$ is the marginal R^2 , and shows how much the first-level effects explain of the variance, and $R^2(c)$ is the conditional R^2 , and shows how much both levels explain. If the $R^2(m)$ is small, and $R^2(c)$ is very big, that means that the second level effects explain a large chunk of the variance, and the model is fitting a lot of noise.

Q: Is it always best practice to use `allfit()` to compare the results of different optimizers, or only when the default optimizer fails to converge?

A:

Explanation versus prediction (W05)

Example question

Q: How does regularisation work?

A: <add your answer>

Please add more questions and answers:

Q: Would you usually need two data sets (two samples) to see if a model you fitted on first sample's data, generalize to second sample's data? Are there any other ways to check, besides just splitting one sample in half?

A:

Q: What is covariance?

A: Covariance is when two variables both increase, decrease, or stay at 0 (co-vary) together. In multi-level modelling, covariance occurs when variables share the same predictors.

Q: What is a cost function? Why do we need it?

A:

Q: What are the indicators of overfitting?

A:

Q: What is the difference between a boolean and a categorical variable?

A: A categorical variable consists of multiple preset, non-numerical values that are distinguished from each other. A boolean variable consists only of two such values, often presented as an either-or variable. A categorical variable must have 3 or more variables while a boolean has exactly 2.

General questions

Example question

Q: What is covariance?

A: <add your answer>

Please add more questions and answers:

Q: How is adding bias good practice for modelling?

A: when dealing with bias and variance, it can be favourable to add bias to a model if doing so reduce variance. higher variance means that the model is less predictable and accurate, a detriment that can be reduced by adding a bias in some circumstances. this all depends on the added bias being known as it can then be accounted for.

Q: what is penalizing??

A:

Q: Could you *theoretically* calculate the betas of a multi-level model by hand like how you can do it with a linear model?

A:

Q: Why are we not supposed to remove NA's in assignment 0?

A: You shouldn't remove trials with NAs for f0mn in assignment 0 because you also remove values from other trials that can improve the model. You should only remove the NAs while plotting in order to limit errors (ggplot can't plot missing values) and reduce visual noise.

Q: What are the differences between bias and variance and how should the two be treated?

A:

Q: what is the point of MSE?

A:

Q: How do you interpret the output of a multilevel model (the different coefficients etc)?

A:

Q for lau: Why cant we get the powerpoint before the lecture? i have a hard time with notetaking and i use more energy on writing notes than listening as i need to write stuff down from the powerpoint. There is not much text on the powerpoint, so later when looking at it, i cannot remember what was said.

A: